

ENTHUSIAST ADVANCE COURSE

PHASE : MEF-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 23-04-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	2	2	3	1	3	1	3	3	1	4	1	3	4	4	1	2	1	2	2	2	1	1	1	1	4	2	1	2	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	1	1	3	2	4	1	1	2	2	1	4	3	1	1	4	2	3	3	1	4	4	2	3	2	2	1	2	3	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	3	4	1	4	2	1	1	3	1	3	2	4	4	2	4	1	3	2	1	4	1	2	2	4	1	3	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	2	3	2	2	2	4	3	4	3	1	1	4	1	3	2	1	1	4	4	1	2	1	1	3	3	4	2	3	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	4	1	2	4	2	3	1	4	4	3	3	2	2	4	3	1	3	2	2	1	4	4	2	1	2	1	1	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	4	4	3	1	2	2	2	2	4	1	3	3	1	3	1	4	4	4	1	3	3	1	3	3	4	3	1	4	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	2	3	3	4	4	2	2	3	3	2	4	1	2	2	1	2	2	4	2										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. **Ans (3)**

Potential difference between two equipotential surfaces A and B

$$V_A - V_B = kq \left(\frac{1}{r_A} - \frac{1}{r_B} \right) = kq \left(\frac{r_B - r_A}{r_A r_B} \right) = \frac{kq t_1}{r_A r_B}$$

$$t_1 = \frac{(V_A - V_B) r_A r_B}{kq} \Rightarrow t_1 \propto r_A r_B$$

Similarly, $t_2 \propto r_B r_C$

Since, $r_A < r_B < r_C$, therefore, $r_A r_B < r_B r_C$, $t_1 < t_2$

2. **Ans (2)**

$\vec{E}$ is maximum at sharp corners while entire conductor is equipotential.

3. **Ans (2)**

Direction of charge flow from inside to outside in conducting concentric hollow metallic spheres.

4. **Ans (3)**

Field lines are smooth curves which originate or terminate perpendicular to the surface of conductor.

5. **Ans (1)**

$$U = -PE \cos \theta \text{ if } \theta = 180^\circ U_{\max} = +PE$$

6. **Ans (3)**

$$a = \frac{qE}{m} \text{ \& by first equation}$$

$$v = u + at$$

$$v = 0 + \frac{qE}{m} t$$

$$K.E. = \frac{1}{2} mv^2 = \frac{1}{2} \times m \left(\frac{qE}{m} t \right)^2$$

$$K.E. = \frac{q^2 E^2 t^2}{2m}$$

7. **Ans (1)**

Total no. of pairs = 15 $\therefore$ 1st option is correct.

8. **Ans (3)**

Area of I-t curve is always equal to charge flow :-

$$\text{So } q_{\text{total}} = \left(\frac{1}{2} \times 5 \times 2 + 4 \times 5 + \frac{1}{2} \times 2(5 + 15) \right) \mu C$$

$$= 45 \mu C$$

13. **Ans (3)**

$$v_d = \frac{i}{A n_c} \quad \text{As } A \uparrow \text{ so } v_d \downarrow \Rightarrow v_p > v_Q$$

14. **Ans (4)**

$$\text{Conductance} = \frac{1}{\text{Resistance}}$$

15. **Ans (4)**

$$\frac{1}{2} m v^2 = \frac{k q_1 q_2}{r}$$

$$\Rightarrow r \propto \frac{1}{v^2}$$

$$\frac{r_1}{r_2} = \left(\frac{v_2}{v_1} \right)^2 = \left(\frac{2v}{v} \right)^2 = 4$$

$$\Rightarrow r_2 = \frac{r_1}{4} = \frac{r}{4}$$

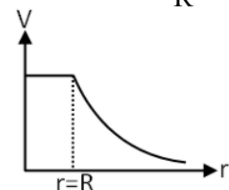
16. **Ans (1)**

It is electric dipole at large distance electric field intensity

$$E = \frac{KP}{R^3} \sqrt{1 + 3\cos^2\theta} \quad \therefore E \propto \frac{1}{R^3}$$

17. **Ans (2)**

$$V_{in} = V_S = \frac{KQ}{R} \quad \text{and} \quad V_{out} = \frac{KQ}{r} \quad (r > R)$$

18. **Ans (1)**

$$\text{Image time } 12 \text{ hrs} - 8 \text{ h } 20 \text{ m} = 3 \text{ h } 40 \text{ m.}$$

19. **Ans (2)**

$$f = \sqrt{xy} \quad (\text{Newtons formula})$$

$$\text{By applying } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

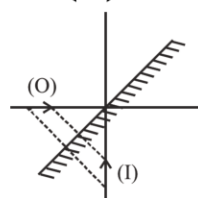
20. **Ans (2)**

$$m = \frac{360}{6} = 6$$

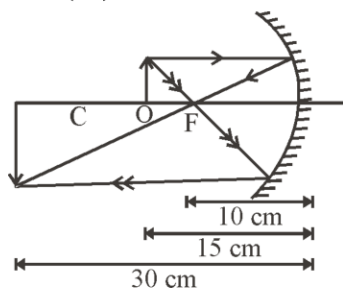
m is even so

$$n = m - 1$$

$$n = 6 - 1 = 5$$

21. **Ans (2)**22. **Ans (1)**

Focal length of mirror doesn't depend on medium.

23. **Ans (1)**

According to New Cartesian sign convention,

Object distance $u = -15 \text{ cm}$

Focal length of a concave lens, $f = -10 \text{ cm}$

Height of the object $h_o = 2.0 \text{ cm}$

According to mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-10} - \frac{1}{-15} \Rightarrow v = -30 \text{ cm.}$$

This image is formed 30 cm from the mirror on the same side of the object. It is a real image.

Magnification of the mirror,

$$m = -\frac{v}{u} = \frac{h_1}{h_o}$$

$$\Rightarrow \frac{-(-30)}{-15} = \frac{h_1}{2} \Rightarrow h_1 = -4 \text{ cm.}$$

Negative sign shows that image is inverted.

The image is real, inverted, of size 4 cm at a distance 30 cm in front of the mirror.

24. **Ans (1)**

The complete image is formed but intensity of image will decrease

Intensity $\propto$ (area of mirror reflecting the light.)

25. **Ans (1)**

$$L_{\text{Image}} = -m^2 L_{\text{Object}}$$

$$|L_{\text{Image}}| = m^2 |L_{\text{Object}}|$$

$$= 4 \times 1 = 4 \text{ mm}$$

$$m = \frac{f}{f - u}$$

$$= \frac{-10}{-10 + 15} = -2$$

26. **Ans (4)**

$$\delta = 180^\circ - 2i$$

$$= 180 - 2 \times 30^\circ$$

$$= 180^\circ - 60^\circ = \boxed{120^\circ}$$

27. Ans (2)

$$\vec{V}_M = 4\hat{i} + 5\hat{j} + 8\hat{k}$$

$$\vec{V}_O = 3\hat{i} + 4\hat{j} + 5\hat{k}$$

$$\perp \text{ direction} \rightarrow \hat{k} \rightarrow V_I = 2(8) - 5 = 11$$

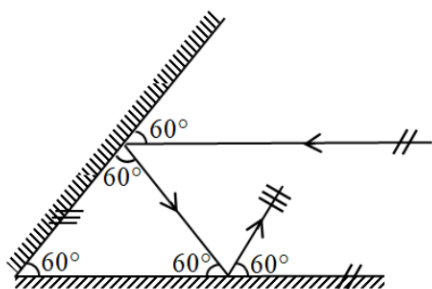
$$\therefore \vec{V}_I = 3\hat{i} + 4\hat{j} + 11\hat{k}$$

28. Ans (1)

$$\frac{360^\circ}{72^\circ} = 5 \text{ (asymmetrical)}$$

$$\therefore \boxed{N = 5}$$

29. Ans (2)



30. Ans (4)

When object is at center of curvature, its real image is formed at the same point.

32. Ans (1)

$f = -20 \text{ cm}$; $m = -2$ (R.O. $\rightarrow$ R.I. $\therefore$ inverted)

$$\Rightarrow \frac{-v}{u} = -2$$

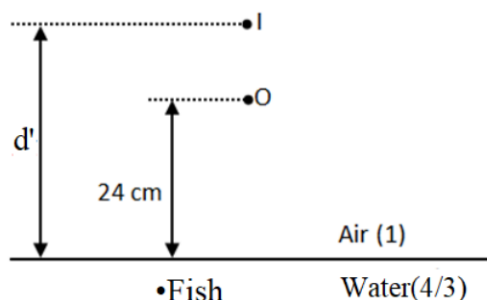
$$\Rightarrow \boxed{v = 2u}$$

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$= \frac{-1}{20} = \frac{1}{2u} + \frac{1}{u}$$

$$\boxed{u = -30 \text{ cm}}$$

33. Ans (1)



$$d' = \frac{d}{\left(\frac{1}{4/3}\right)} = \frac{4d}{3}$$

$$\Rightarrow d' = \frac{4}{3} \times 24 \text{ cm} = \boxed{32 \text{ cm}}$$

34. Ans (3)

$$n = \frac{c}{v} \Rightarrow v = \frac{c}{n} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ m/s}$$

$$\text{Now, } t = \frac{4 \text{ mm}}{v} = \frac{4 \times 10^{-3} \text{ m}}{2 \times 10^8 \text{ m/s}} = \boxed{2 \times 10^{-11} \text{ s}}$$

35. Ans (2)

$$n = \frac{c}{v} = \frac{\lambda_{\text{vacuum}} \times f}{\lambda_{\text{medium}} \times f}$$

$$\Rightarrow n = \frac{\lambda_{\text{vacuum}}}{\lambda_{\text{medium}}} = \frac{6000}{4000} = \boxed{1.5}$$

SECTION-B

36. Ans (4)

$$V = \frac{Q}{4\pi\epsilon_0 \times 1} - \frac{Q}{4\pi\epsilon_0 \times 2} + \frac{Q}{4\pi\epsilon_0 \times 2^2} \dots \infty$$

$$= \frac{Q}{4\pi\epsilon_0} \left[1 - \frac{1}{2} + \frac{1}{2^2} - \frac{1}{2^3} + \dots \infty \right]$$

$$= \frac{Q}{4\pi\epsilon_0} \times \frac{1}{1 - (-1/2)} = \frac{Q}{4\pi\epsilon_0} \times \frac{2}{3} = \frac{Q}{6\pi\epsilon_0}$$

37. Ans (1)

for $r < a$, $E = 0$

$$\text{for } a < r < b, \vec{E} = \frac{kQ}{r^2} \hat{r}$$

$$\text{for } r > b, \vec{E} = -\frac{kQ}{r^2} \hat{r}$$

38. Ans (1)

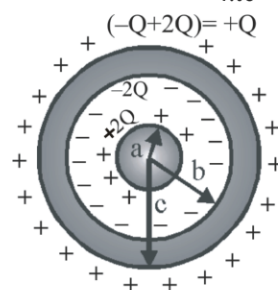
After redistribution of charge by induction,

Net charge on inner surface of shell $= -2Q$

$$\text{Charge density} = \frac{-2Q}{4\pi b^2}$$

Net charge on outer surface of shell $= -Q + 2Q = +Q$

$$\text{Charge density} = \frac{+Q}{4\pi c^2}$$



40. Ans (2)

$$i = qf$$

$$= 6.6 \times 10^{15} \times 1.6 \times 10^{-19}$$

$$= 10.56 \times 10^{-4}$$

$$\approx 1 \text{ mA}$$

44. **Ans (1)**

$$m = -16$$

$$u = -x$$

$$v = -(120 + x)$$

$$m = \frac{-v}{u}$$

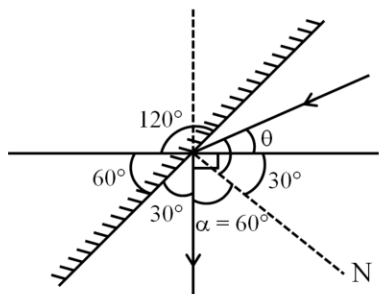
$$-16 = \frac{120 + x}{-x}$$

$$15x = 120$$

$$x = 8\text{cm}$$

45. **Ans (1)**

$$\delta = t \left(1 - \frac{1}{\mu} \right) \text{ in the direction of propagation (away)}$$

46. **Ans (4)**

angle of incidence = angle of reflection

$$\theta + 30^\circ = \alpha$$

$$\theta = \alpha - 30^\circ$$

$$= 60^\circ - 30^\circ$$

$$= 30^\circ$$

47. **Ans (2)**

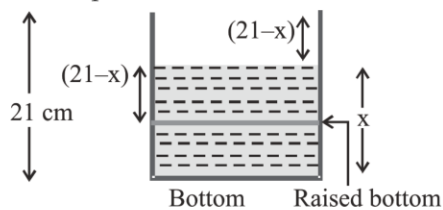
$$\text{Raise} = 3 \left(1 - \frac{1}{3/2} \right) = 1\text{cm}$$

Object is at C

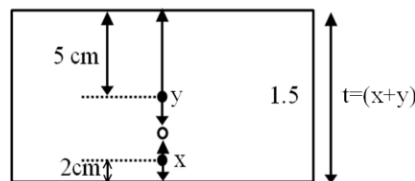
Image is at C

48. **Ans (3)**

To see the container half-filled from top, water should be filled up to height x so that bottom of the container should appear to be raised upto height $(21-x)$. As shown in figure apparent depth $h' = (21-x)$

Real depth $h = x$ 

$$\therefore \mu = \frac{h}{h'} \Rightarrow \frac{4}{3} = \frac{x}{21-x} \Rightarrow x = 12\text{cm}$$

49. **Ans (3)**

$$5 = \frac{y}{1.5}$$

$$\Rightarrow y = 7.5\text{ cm}$$

$$2 = \frac{x}{1.5}$$

$$\Rightarrow x = 3\text{ cm}$$

$$\therefore t = x + y = 10.5\text{cm}$$

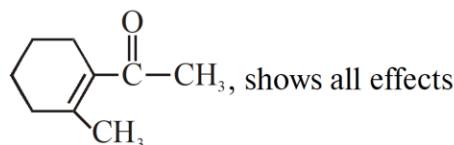
50. **Ans (1)**

$$n_{\text{relative}} = \frac{n_{\text{Quartz}}}{n_{\text{Glycerine}}} = \frac{V_{\text{Glycerine}}}{V_{\text{Quartz}}} = \frac{9/4 \times 10^8}{1.5 \times 10^8} = 1.5$$

$$\text{Now, shift} = t \left(1 - \frac{1}{n_{\text{rel}}} \right)$$

$$= 18 \left(1 - \frac{1}{1.5} \right)$$

$$= 6\text{cm}$$

SUBJECT : CHEMISTRY**SECTION-A**59. **Ans (3)**77. **Ans (4)**

$$\frac{P^\circ - P}{P^\circ} = x_{\text{solute}} \text{ or } \frac{P^\circ - P}{P^\circ} = \frac{n}{n + N}$$

$$\Rightarrow \frac{0.1}{0.1 + 9.9} \Rightarrow 0.01; P = 0.99 P^\circ$$

SUBJECT : BIOLOGY-I**SECTION-A**111. **Ans (1)**

NCERT Pg#140

112. **Ans (2)**

NCERT Pg#140

113. **Ans (1)**

NCERT XII Page No136

116. **Ans (3)**

NCERT-XII, Pg. # 141

117. **Ans (4)**
Module Pg. # 123,124
118. **Ans (2)**
NCERT-XII, Pg. # 140
119. **Ans (3)**
NCERT-XII, Pg. # 140, 141
120. **Ans (3)**
NCERT-XII, Pg. # 140
121. **Ans (4)**
NCERT-XII, Pg. # 135, 137, 140
122. **Ans (4)**
NCERT-XII, Pg. # 138
123. **Ans (1)**
NCERT-XII, Pg. # 130
124. **Ans (2)**
Module, Pg. # 119 Golden key points
125. **Ans (4)**
Module- Pg. # 112
126. **Ans (2)**
Module- Pg. # 115
127. **Ans (3)**
NCERT-XII, Pg. # 136, 137
129. **Ans (4)**
Module- Pg. # 118
130. **Ans (4)**
Module- Pg. # 115, 116
131. **Ans (3)**
NCERT-XII, Pg. # 131, 132
132. **Ans (3)**
NCERT-XII, Pg. # 134
134. **Ans (2)**
NCERT-XII, Pg. # 135
135. **Ans (4)**
NCERT-XII, Pg. # 142

SECTION-B

136. **Ans (3)**
Module
141. **Ans (1)**
NCERT Page No. 135

146. **Ans (2)**
NCERT-XII, Pg. # 141
147. **Ans (1)**
NCERT-XII, Pg. # 141
148. **Ans (1)**
NCERT-XII, Pg. # 138
150. **Ans (3)**
Module- Pg. # 121,122,123

SUBJECT : BIOLOGY-II**SECTION-A**

155. **Ans (1)**
NCERT (XIIth) Pg. # 22
156. **Ans (2)**
NCERT XII Pg. # 25
158. **Ans (2)**
NCERT (XII) Pg # 26
159. **Ans (2)**
NCERT Page#26
160. **Ans (4)**
NCERT (XII) Pg. # 30 (E), (31) H
161. **Ans (1)**
NCERT XIIth Pg.#25(E), 26(H)
164. **Ans (1)**
NCERT (XII) Pg. # 31 (E), 33 (H)
165. **Ans (3)**
NCERT (XII) Pg. # 29 last para
168. **Ans (4)**
NCERT-XII Page # 28, 23, 25

SECTION-B

186. **Ans (4)**
NCERT Pg # 34
188. **Ans (2)**
NCERT XII pg. 21 Fig. 2.2 (b)
189. **Ans (3)**
NCERT XIIth Pg.#34(E), 36(H)
190. **Ans (3)**
NCERT XII Pg. # 34
191. **Ans (2)**
NCERT (XII) Pg. # 29
192. **Ans (4)**
NCERT XII Pg # 32